

Piola identities and boundary integrals

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Scope

Mollification, the Piola identity and determinant telescoping give integral identities for compact perturbations. Strong local convergence then passes to Lipschitz maximal minors, whose integrals are determined by boundary agreement.

10 direct Cards + 1 reused prerequisite = 11 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [Boundary agreement determines maximal-minor integrals](#)
- Key result: [The divergence-free cofactor identity](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#) [5](#)

Level 0 (4 Cards)

Level 0

Mollification by a normalized smooth kernel

Defines convolution with a nonnegative smooth kernel of integral one.

`MathlibAnnex.Mollification.mollify`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Strong local convergence of mollified derivatives](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A cofactor row by row replacement

Fixes the cofactor convention used in the divergence and determinant identities.

`MathlibAnnex.Piola.cofactorRow`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The divergence-free cofactor identity](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Zero integral of a compactly supported divergence

Makes all faces in a box divergence formula vanish by placing the support strictly inside the box.

`MathlibAnnex.Piola.integral_coordinateDivergence_eq_zero_of_contDiff_hasCompactSupport`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A determinant difference bound in the sup operator norm

Controls determinant variation by replacing one matrix row at a time.

`MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Determinant integrals under strong \$L^n\$ convergence](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (3 Cards)

Level 1

Strong local convergence of mollified derivatives

Approximates the derivative in local L^n , where n is the dimension of the domain.

```
MathlibAnnex.Mollification.tendsto_elpNorm_fderiv_mollify_sub
```

Immediate Card prerequisites: [Mollification by a normalized smooth kernel](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The divergence-free cofactor identity

Cancels the Hessian terms that arise when differentiating a cofactor row.

```
MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero
```

Immediate Card prerequisites: [A cofactor row by row replacement](#)

Used by in this scope: [A component flux gives a determinant difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Determinant integrals under strong L^n convergence

Uses Hölder's inequality to turn strong L^n convergence of matrix fields into convergence of their determinant integrals.

```
MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn
```

Immediate Card prerequisites: [A determinant difference bound in the sup operator norm](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (1 Card)

Level 2

A component flux gives a determinant difference

Expresses the change from one output-coordinate perturbation as a divergence.

`MathlibAnnex.Piola.divergence_componentFlux_eq_det_sub`

Immediate Card prerequisites: [The divergence-free cofactor identity](#)

Used by in this scope: [Smooth compact perturbations preserve the determinant integral difference](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (1 Card)

Level 3

Smooth compact perturbations preserve the determinant integral difference

Builds a finite telescope from single-output perturbations with compactly supported fluxes.

`MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff`

Immediate Card prerequisites: [A component flux gives a determinant difference](#) · [Zero integral of a compactly supported divergence](#)

Used by in this scope: [Maximal-minor integral differences for Lipschitz perturbations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (1 Card)

Level 4

Maximal-minor integral differences for Lipschitz perturbations

Passes the smooth compact-perturbation identity to Lipschitz maps through local strong convergence.

```
MathlibAnnex.NullLagrangian.integral_maximalMinor_
fderiv_add_sub_eq_zero_of_lipschitzWith
```

Immediate Card prerequisites: [Determinant integrals under strong \$L^n\$ convergence](#) · [Strong local convergence of mollified derivatives](#) · [Smooth compact perturbations preserve the determinant integral difference](#)

Used by in this scope: [Boundary agreement determines maximal-minor integrals](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5 (1 Card)

Level 5

Boundary agreement determines maximal-minor integrals

Turns pointwise equality on a seminorm sphere into a compactly supported Lipschitz perturbation.

```
MathlibAnnex.NullLagrangian.integral_maximalMinor_
eq_of_pointwise_boundary_eq
```

Immediate Card prerequisites: [Maximal-minor integral differences for Lipschitz perturbations](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)